

A path certificate for Wilf’s conjecture and verification up to multiplicity 23

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Abstract

We give a lower bound for the Wilf number of a numerical semigroup in terms of a finite downset of Apéry normal forms and the residues of its minimal generators. The bound is a weighted longest-path certificate and is uniform in the generator heights. A discrete summation identity and a carry estimate yield the shape criterion $W(S) \geq R_0(B) - D(B)$. Two independent exhaustive enumerations, each rerun in full, verify this criterion on every required cyclic tile of multiplicities 20 through 23, with one exception at multiplicity 21. That exception is certified directly at all maximal targets in its unit orbit. Combining these calculations with established cases proves Wilf’s conjecture for all numerical semigroups of multiplicity at most 23, without a bound on genus or conductor.

1 Introduction

A numerical semigroup is a submonoid $S \subseteq \mathbb{N} = \{0, 1, 2, \dots\}$ with finite complement. For $S \neq \mathbb{N}$, write m for its multiplicity, e for the cardinality of its minimal generating set, and c for its conductor. Put $n = |S \cap [0, c)|$. Wilf’s conjecture asks whether

$$W(S) := en - c \geq 0 \tag{1}$$

always holds [1].

Bruns, García-Sánchez, O’Neill and Wilburne established the conjecture for $m \leq 18$ using a polyhedral algorithm [4]. Kliem and Stump extended this to $m = 19$ [5]. The cases $e \leq 3$ are classical [2, 3]; Eliahou proved the conjecture whenever $3e \geq m$ [6]. Genus-based computation has verified it through genus 100 [8]. The August 2026 account [9] still lists $m \leq 19$ as the fixed-multiplicity bound.

Here we use finite downsets of exponent vectors representing the Apéry set. Downsets, also viewed as sets of monomials closed under division, already play a role in the literature; see, for example, [7]. Our purpose is to give a self-contained path certificate on such a downset and apply it to a finite census. The construction is uniform in the integer heights of the generators, so a finite calculation addresses infinitely many semigroups.

Theorem 1.1 (Computer-assisted). *Every numerical semigroup of multiplicity $m \leq 23$ satisfies Wilf’s conjecture.*

For $20 \leq m \leq 23$, the only cases not supplied by $e \leq 3$ and $3e \geq m$ are

m	embedding dimensions requiring computation
20, 21	4, 5, 6
22, 23	4, 5, 6, 7.

These are the fourteen rows of Table 1. No assertion for unrestricted multiplicity follows from this finite computation.

2 Apéry normal forms and a summation identity

Let

$$S = \langle m, g_1, \dots, g_d \rangle, \quad e = d + 1,$$

be a minimal generating set. Each $g_j > m$ has the form $g_j = a_j + mh_j$, with $1 \leq a_j < m$ and integer $h_j \geq 1$. The a_j are distinct: two generators in the same residue class would make the larger one decomposable. Also $\gcd(m, a_1, \dots, a_d) = 1$.

The Apéry set

$$\text{Ap}(S, m) = \{s \in S : s - m \notin S\}$$

contains one element in each residue class modulo m . For each of its elements, select the lexicographically least factorization over $g_1, \dots, g_d$. Let $B \subseteq \mathbb{N}^d$ be the set of selected exponent vectors, and put $w(b) = \sum_j g_j b_j$.

Lemma 2.1. *The set B is a downset of cardinality m , contains the origin and all unit vectors, and the map $\phi(b) = \sum_j a_j b_j \pmod{m}$ is bijective on B .*

Proof. If $b \in B$ and $v \leq b$ coordinatewise, then $w(v) \in \text{Ap}(S, m)$: otherwise $w(b) - m = (w(v) - m) + w(b - v)$ would belong to S . If a factorization of $w(v)$ were lexicographically smaller than v , adding $b - v$ would give a smaller factorization of $w(b)$. Thus $v \in B$. Each g_j lies in $\text{Ap}(S, m)$, since otherwise $g_j = m + (g_j - m)$ would be decomposable, and by minimality $\mathbf{e}_j$ is its only factorization over $g_1, \dots, g_d$. Hence B contains the unit vectors, and the origin, the factorization of 0. Distinct elements of $\text{Ap}(S, m)$ have distinct selected factorizations, so $|B| = m$. Finally $w(b) \equiv \phi(b) \pmod{m}$, and w maps B onto $\text{Ap}(S, m)$, which meets every residue class modulo m exactly once. Therefore ϕ is bijective on B . $\square$

Definition 2.2. A *nondegenerate cyclic tile* of size m in $\mathbb{N}^d$ is a downset B of cardinality m containing the origin and all unit vectors, together with labels $a \in \{1, \dots, m - 1\}^d$ such that $\phi(b) = a \cdot b \pmod{m}$ is bijective on B .

The census includes every such tile, whether or not it arises from a numerical semigroup. This larger class suffices by Lemma 2.1. Write $|b| = \sum_j b_j$ and define the endpoint weight

$$\mu(b) = \sum_{j: b + \mathbf{e}_j \notin B} (b_j + 1). \quad (2)$$

In particular, $\mu(b) \geq 0$.

Lemma 2.3 (Discrete summation). *For every function $f : B \rightarrow \mathbb{R}$,*

$$\sum_{b \in B} \mu(b) f(b) = d \sum_{b \in B} f(b) + \sum_{b \in B} \sum_{j: b_j > 0} b_j (f(b) - f(b - \mathbf{e}_j)). \quad (3)$$

Consequently

$$\sum_b \mu(b) = dm, \quad \sum_b \mu(b) b_j = (d + 1) \sum_b b_j \quad (1 \leq j \leq d). \quad (4)$$

Proof. Expand the left side of (3) as

$$\sum_b (|b| + d)f(b) - \sum_{\substack{b,j \\ b+\mathbf{e}_j \in B}} (b_j + 1)f(b).$$

In the second sum, substitute $v = b + \mathbf{e}_j$. The downset property identifies the indexing set with all (v, j) satisfying $v_j > 0$. This gives (3). Apply it to $f = 1$ and $f(b) = b_j$. $\square$

3 The path certificate

For a cyclic tile, put $A(b) = a \cdot b$. At a target $t \in \{0, \dots, m-1\}$, set

$$l_t = m - 1 - t, \quad N_t(b) = \left\lfloor \frac{A(b) + l_t}{m} \right\rfloor, \quad (5)$$

$$\Psi_t(b) = |b| + N_t(b), \quad E_t(b) = \max_{z \in B, z \geq b} (\Psi_t(z) - \Psi_t(b)). \quad (6)$$

The maximum in (6) is attained at a coordinatewise maximal cell: Ψ_t increases strictly at each upward unit step. Define

$$p_t = l_t + \sum_b \mu(b)N_t(b) - (d+1) \sum_b N_t(b), \quad (7)$$

$$R_t = \sum_b \mu(b)E_t(b), \quad \text{PC}_t = p_t + R_t. \quad (8)$$

All these quantities are integers computed from the finite labelled tile.

For a genuine semigroup, let $\beta = \max \text{Ap}(S, m) = c + m - 1$ and let $y \in B$ be its selected factorization. The cell y is maximal, since the generators are positive. Write $r = \beta \pmod{m}$, $q = \lfloor \beta/m \rfloor$, and

$$\ell_b = \left\lfloor \frac{\beta - w(b)}{m} \right\rfloor \geq 0.$$

Proposition 3.1 (Exact endpoint identity). *With the notation above,*

$$W(S) = p_r + \sum_b \mu(b)\ell_b. \quad (9)$$

Proof. There are exactly ℓ_b members of $w(b) + m\mathbb{N}$ below $c = \beta - m + 1$. Thus $n = \sum_b \ell_b$. Since $c = mq - l_r$ and $w(b) = A(b) + mh \cdot b$,

$$\ell_b = q - h \cdot b - N_r(b).$$

It follows that

$$W = dmq - (d+1) \sum_b h \cdot b - (d+1) \sum_b N_r(b) + l_r.$$

The two identities in (4) give

$$\sum_b \mu(b)\ell_b = dmq - (d+1) \sum_b h \cdot b - \sum_b \mu(b)N_r(b).$$

Subtract and use (7). $\square$

Theorem 3.2 (Path bound). *For every numerical semigroup with the normal-form tile above,*

$$\ell_b \geq E_r(b) \quad (b \in B), \quad W(S) \geq \text{PC}_r. \quad (10)$$

Proof. Let $z \in B$ with $z \geq b$. The formula for ℓ in the proof of Proposition 3.1 and $h_j \geq 1$ give

$$\ell_b - \ell_z = h \cdot (z - b) + N_r(z) - N_r(b) \geq |z| - |b| + N_r(z) - N_r(b).$$

As $\ell_z \geq 0$, maximizing over z proves the first inequality. Multiply it by the nonnegative weights $\mu(b)$ and apply Proposition 3.1. $\square$

The path bound uses only $h_j \geq 1$ and $\ell_z \geq 0$; it requires no upper bound on the heights.

4 A criterion depending only on the shape

Define

$$\eta(b) = \max_{z \in B, z \geq b} (|z| - |b|), \quad R_0(B) = \sum_b \mu(b) \eta(b), \quad D(B) = \sum_b (|b| - 2)_+. \quad (11)$$

Here $x_+ = \max(x, 0)$; the subscript in R_0 indicates a shape quantity, not the target $t = 0$ in R_t .

Lemma 4.1 (Carry estimate). *For every cyclic tile and every target t , one has $p_t \geq -D(B)$.*

Proof. Put $\rho_t(b) = (A(b) + l_t) \pmod{m}$ and $s_t(b) = \sum_{j: a_j > \rho_t(b)} b_j$. If $b_j > 0$, subtracting a_j shows that

$$N_t(b) - N_t(b - \mathbf{e}_j) = \mathbf{1}_{\{a_j > \rho_t(b)\}}.$$

By (3), $p_t = l_t - \sum_b (N_t(b) - s_t(b))$. Since ρ_t is a permutation of the residues, precisely l_t of its values exceed t . Therefore

$$p_t = - \sum_b F_t(b), \quad F_t(b) = N_t(b) - s_t(b) - \mathbf{1}_{\{\rho_t(b) > t\}}. \quad (12)$$

We claim $F_t(b) \leq |b| - 2$ when $|b| \geq 2$, and $F_t(b) \leq 0$ when $|b| \leq 1$. For $b = 0$ this is immediate. For a unit vector, $N_t(b) \in \{0, 1\}$; if it is 1, then $\rho_t(b) < a_j$ and $s_t(b) = 1$, proving the claim.

Now write $k = |b| \geq 2$, $s = s_t(b)$, $\rho = \rho_t(b)$, $N = N_t(b)$, and $l = l_t$. If $s = k$, then $Nm + \rho = A(b) + l \leq k(m - 1) + l < (k + 1)m$, so $N \leq k$ and $F_t(b) \leq 0$. If $s < k$, the $k - s$ noncarrying label occurrences are at most ρ , and the remaining s are at most $m - 1$. Hence

$$Nm \leq (k - s - 1)\rho + s(m - 1) + l \leq (k - 1)(m - 1) + l < km.$$

Thus $N \leq k - 1$. If $s \geq 1$, or if $N \leq k - 2$, the claim follows. The remaining possibility is $s = 0$ and $N = k - 1$. In that case $(k - 1)(m - \rho) \leq l$. If $\rho \leq t$, the left side is at least $(k - 1)(m - t) > m - 1 - t = l$, a contradiction. Consequently $\rho > t$, and $F_t(b) = k - 2$. Summing the claim in (12) proves the lemma. $\square$

Lemma 4.2. *For every cyclic tile and target, $R_t \geq R_0(B)$.*

Proof. The labels are positive, so $N_t(z) \geq N_t(b)$ for $z \geq b$. It follows that $E_t(b) \geq \eta(b)$. Sum with weights $\mu(b)$. $\square$

Corollary 4.3 (Shape certificate). *Every numerical semigroup satisfies*

$$W(S) \geq \text{PC}_r \geq R_0(B) - D(B). \quad (13)$$

In particular, $R_0(B) \geq D(B)$ suffices for Wilf's conjecture, for every realization of the labelled tile and every genus.

The criterion is sufficient; no universal assertion $R_0(B) \geq D(B)$ is made. The next section includes a cyclic tile where that inequality fails and the path certificate still succeeds.

5 The finite verification

5.1 Finite candidate sets and exhaustive enumeration

For any nondegenerate downset of size m , every nonzero $b \in B$ obeys

$$|b| \leq m - d, \quad \prod_{j=1}^d (b_j + 1) + \#\{j : b_j = 0\} \leq m. \quad (14)$$

Indeed, a monotone path to b contains $|b| + 1$ cells and exactly one unit vector, so adjoining the other $d - 1$ units gives the first bound. The box $[0, b]$ has the displayed product as its cardinality; the units in zero coordinates lie outside that box, giving the second bound. The origin is included separately. These inequalities give an explicit finite candidate set containing every possible tile.

Both production programs enumerate strictly increasing label vectors with entries in $\{1, \dots, m - 1\}$ and with $\gcd(m, a_1, \dots, a_d) = 1$. For a unit $u \in (\mathbb{Z}/m\mathbb{Z})^\times$, multiplying labels by u preserves bijectivity. Sorting the resulting labels is a simultaneous coordinate permutation of the tile. One lexicographically least sorted label vector is chosen from each unit orbit. The orbit weight is the number of distinct sorted label vectors in that orbit, which may be smaller than $\varphi(m)$. The shape quantities R_0 and D are invariant under coordinate permutation. These reductions therefore preserve all required shapes.

The first enumerator, archived as `tiles_enum.py`, constructs every cell whose box has injective residue labels. Such cells form a downset, and each has box volume at most m , so the construction is finite. Every cell of a cyclic tile must occur in this pool. The search assigns one cell to each residue. Choosing a candidate forces all cells in its box; a branch is rejected only when a residue would be assigned two different cells or an unassigned residue has no consistent candidate. It chooses the next residue deterministically with the fewest consistent candidates. For any fixed valid tile, its assigned cell is always available along the branch corresponding to that tile, so the search finds every tile. The deterministic branching and forced assignments also give a unique branch for each completed tile. Finally the nondegeneracy condition is imposed. The production calls have no tile cap or degree cutoff.

The second enumerator uses (14), orders candidate cells by degree and then lexicographically, and extends the forced origin and units by an increasing sequence. A cell may be added only when its immediate parents have already been selected and its residue is unused. Every downset has exactly one such increasing sequence. A branch is pruned when an unused residue has no later candidate, or every later candidate of that residue has an already excluded parent. Both tests are necessary conditions for an extension and cannot discard a tile. This search shares no exact-cover enumeration code with the first one.

5.2 Completed census rows

Table 1 contains exactly the cases needed beyond the established theorems. The weighted count is the sum of label-orbit weights, not the number of representatives actually visited. In fresh full runs, the two programs give identical weighted counts, minima of $R_0 - D$ and exception counts on every row; the representative counts are from the first program, as the second does not report them. The representative count includes every tile at the chosen label vectors; it does not take a further quotient by their stabilizers.

Write $H = \sum_b (|\text{supp}(b)| - 1)_+$. An earlier version of the first census evaluated only $R_0 - \min(H, D)$. Nonnegative values of that margin alone do not establish the criterion used here. Both

m	$d = e - 1$	Representatives	Weighted tiles	$\min(R_0 - D)$	Exceptions
20	3	1,391	10,244	4	0
20	4	25,619	187,000	12	0
20	5	328,112	2,492,114	14	0
21	3	1,383	15,846	-2	1
21	4	30,133	331,341	16	0
21	5	444,166	5,195,104	11	0
22	3	1,763	17,265	0	0
22	4	41,611	397,790	14	0
22	5	664,133	6,574,135	12	0
22	6	8,169,530	81,030,042	19	0
23	3	1,487	32,714	1	0
23	4	37,699	795,993	16	0
23	5	670,311	14,746,842	18	0
23	6	9,460,136	206,425,967	16	0
Total		19,877,474	318,252,397		1

Table 1: The fourteen required census rows. Exceptions count representatives with $R_0 < D$.

programs now record $R_0 - D$ directly; these minima and the direct treatment below supply precisely the evidence needed for Corollary 4.3. The present proof does not require a bound involving H .

5.3 The exception at multiplicity 21

Both programs report the same sole exceptional representative in Table 1. It has $a = (1, 4, 5)$ and

$$B = \{(0, j, k) : j, k \geq 0, j + k \leq 4\} \cup \{(i, 0, k) : 1 \leq i \leq 3, k \geq 0, i + k \leq 3\}. \quad (15)$$

It has $|B| = 21$, $R_0(B) = 15$ and $D(B) = 17$. The eight maximal cells are

$$\{(0, j, 4 - j) : 0 \leq j \leq 4\} \cup \{(i, 0, 3 - i) : 1 \leq i \leq 3\}.$$

For each of the twelve units $u \pmod{21}$, replace a by $a^{(u)} = (u, 4u, 5u) \pmod{21}$, with residues in $\{1, \dots, 20\}$, and evaluate (8) at each maximal target $t = a^{(u)} \cdot z \pmod{21}$. The minima over the eight targets are

u	1	2	4	5	8	10	11	13	16	17	19	20
$\min_t \text{PC}_t$	15	22	31	23	29	23	23	22	25	22	23	23.

Thus all 96 certificates are nonnegative. Simultaneous coordinate permutation changes none of their values. Every genuine realization of this orbit has a maximal Apéry target, so Theorem 3.2 certifies it.

Proof of Theorem 1.1. The semigroup $\mathbb{N}$ is immediate. The case $m \leq 19$ follows from [4, 5]. For $20 \leq m \leq 23$, apply the established results when $e \leq 3$ or $3e \geq m$. In every remaining case Lemma 2.1 gives a tile included in Table 1. Except for (15), it satisfies $R_0 \geq D$, so Corollary 4.3 applies. The preceding calculation certifies the exceptional orbit. This exhausts all cases. $\square$

6 Reproducibility and scope

The computational supplement, supplied as ancillary files with the arXiv source submission, contains the two production algorithms, the outputs of their full reruns, the earlier archived row summaries and terminal logs, a SHA-256 manifest, and `verify.py`. Its default checks recompute the extrema saved in 99 archived records, compare the fourteen required rows of the archived and rerun outputs, check that both reruns report the same single exceptional tile, evaluate the 96 exceptional certificates, and verify the endpoint and path identities on eight retained genuine semigroups and fifty seeded examples. Its `--smoke` option also performs fourteen fresh small enumerations through $m = 9$, compares the complete tile sets of an exact-cover and a linear-extension search for each label vector, and tests the carry estimate at every target on those tiles.

The independent program was compiled with GCC 13.3.0 (`-O3 -std=gnu11`) and run under Ubuntu 24.04 in the Windows Subsystem for Linux; the exact-cover census was run with CPython 3.14.5 under Windows 11. The supplement records the commands, source hashes, run times and outputs. All fourteen C runs recorded exit code zero. Python exit codes were not captured; each run produced its final summary record, written after enumeration, and left standard error empty. These reruns, rather than the archived logs, are the evidence for Table 1. The archived outputs agree with them on every row, but some were produced by source revisions not retained in the supplement.

Two further checks share no code with the production programs. `shape_first_d3.py` enumerates every downset of size m in $\mathbb{N}^3$ containing the units, counts its valid labellings directly, and reproduces the four rows with $d = 3$, including the exception and its certificates. `genus_sweep.py` checks Lemmas 2.1, 2.3, 4.1 and 4.2, Proposition 3.1 and Theorem 3.2 on all 258,582 numerical semigroups of genus at most 22, testing the carry and height bounds at every target. Fourteen of these semigroups have $R_0(B) < D(B)$; the path certificate covers each of them. Instructions for reproducing the computations and checking the saved evidence are included in the supplement.

The theorem is uniform in genus and generator height. The certificate itself applies in every dimension, but the finite census establishes its required sign only in the stated range. Extending that range or proving positivity for general cyclic tiles requires further work.

Computer and AI assistance. This is a computer-assisted proof: its finite verification uses the exhaustive enumerations described above. AI tools also assisted with mathematical exploration, proof development, code preparation and manuscript drafting and review. The retained sources and outputs make the computational evidence available for inspection and reproduction. The author is responsible for the mathematical claims and the submitted manuscript.

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